Incentivizing Retail Traders: Evidence from Daily High-Water Marks on a Social Trading Platform

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Technology is Changing How Retail Investors Trade

- Fintech has changed how consumers interact with financial markets
  - Spending (D'Acunto et al. 2019b) and saving (Gargano and Rossi 2024)
  - Robo-advisors help investors diversify (D'Acunto et al. 2019a)
- Platform design shapes the decisions retail investors make
  - 'Gamified' interfaces alter investor attention (Barber et al. 2022; Tan 2021)
- *A growing and relatively unexplored innovation: social trading platforms*

Social Trading: Retail Traders as Fund Managers

- Retail **traders** make their investment decisions public
  - Retail **investors** buy an exchange-traded product that tracks the trader's portfolio
  - Retail traders effectively become fund managers
- The platform pays traders a performance bonus
  - A fraction of the fund's return above its previous high close
  - Earned any day the portfolio closes above that high, and the threshold resets immediately
- *An institutional high water mark contract, but paid daily to retail traders* (Ma et al. 2019; Aragon and Nanda 2012)

How Could Daily Incentives Affect Retail Traders?

- Alignment
  - Performance pay aligns trader and investor interests, potentially mitigating retail biases
    - (Jensen and Meckling 1976; Holmstrom 1979)
- Or amplification
  - Extreme frequency amplifies loss aversion, excessive trading, and short-term thinking
    - (Odean 1998, 1999; Dorn et al. 2015; Delfabbro et al. 2021)
- **This paper: How does a daily high water mark bonus change the behavior and performance of retail traders?**

Data and Methodology

- Data
  - **Social trading platform:** every trade, daily fund prices, and bonus days from 2018 to 2022
  - **Sample:** 5,147 funds and 3.6 million fund-day observations
  - **CRSP and LSEG:** returns of the underlying assets
  - *Increasing our coverage with German exchange data*
- Methodology
  - Regression discontinuity around the bonus threshold
  - Differences-in-differences around each bonus (de Chaisemartin and D'Haultfoeuille 2026)
  - January 1st reset of the high water mark
  - Counterfactual static portfolio where traders do not trade at all

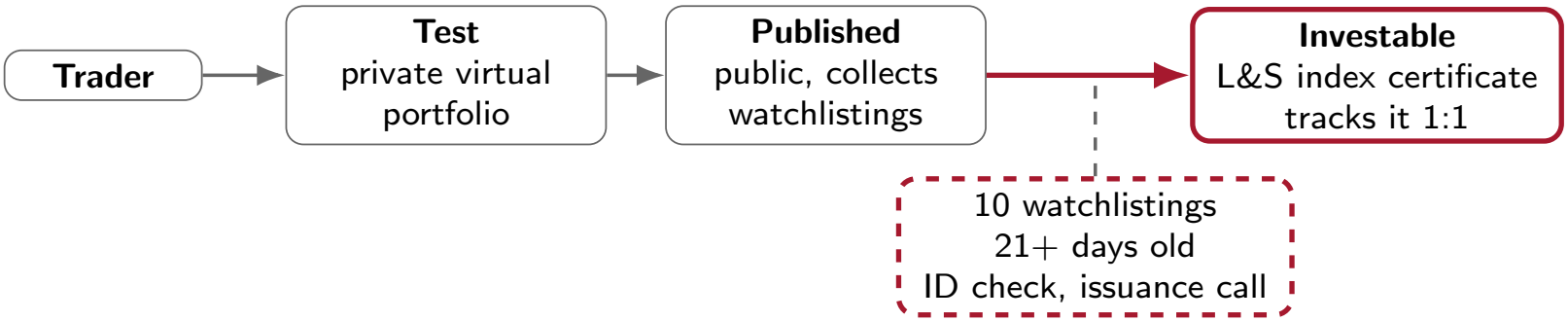
Preview of the Main Results

- On days a bonus is achieved
  - Returns spike 1.6 percentage points above the previous day
    - Immediately fall back to pre-bonus levels
  - Odds of trading are 15% higher just above the bonus threshold
  - Traders sell recent winners to lock in the gain, but miss further run up
- Doing nothing would be better: 63% of funds would earn higher alpha and 57% of fund-years would earn more bonuses
- Consistent with daily incentives amplifying short-term focus rather than aligning interests

Contribution

- Performance incentives are well studied for institutional managers (Brown et al. 1996; Massa and Patgiri 2009; Agarwal et al. 2009; Ray 2009); pay can align interests (Jensen and Meckling 1976; Holmstrom 1979)
  - Daily HWM amplifies short-term focus among retail traders
- Fintech platform design can help (D'Acunto et al. 2019a; Rossi and Utkus 2021; Gargano and Rossi 2024) or hurt users (Barber et al. 2022; Tan 2021)
  - Wall Street style incentives backfire at a daily frequency
- Retail and social trading (Barber and Odean 2000; Petajisto 2013; Anginer et al. 2024; Oehler et al. 2016; Roder and Walter 2019; Oehler and Schneider 2023)
  - Retail–professional gap persists even with performance pay

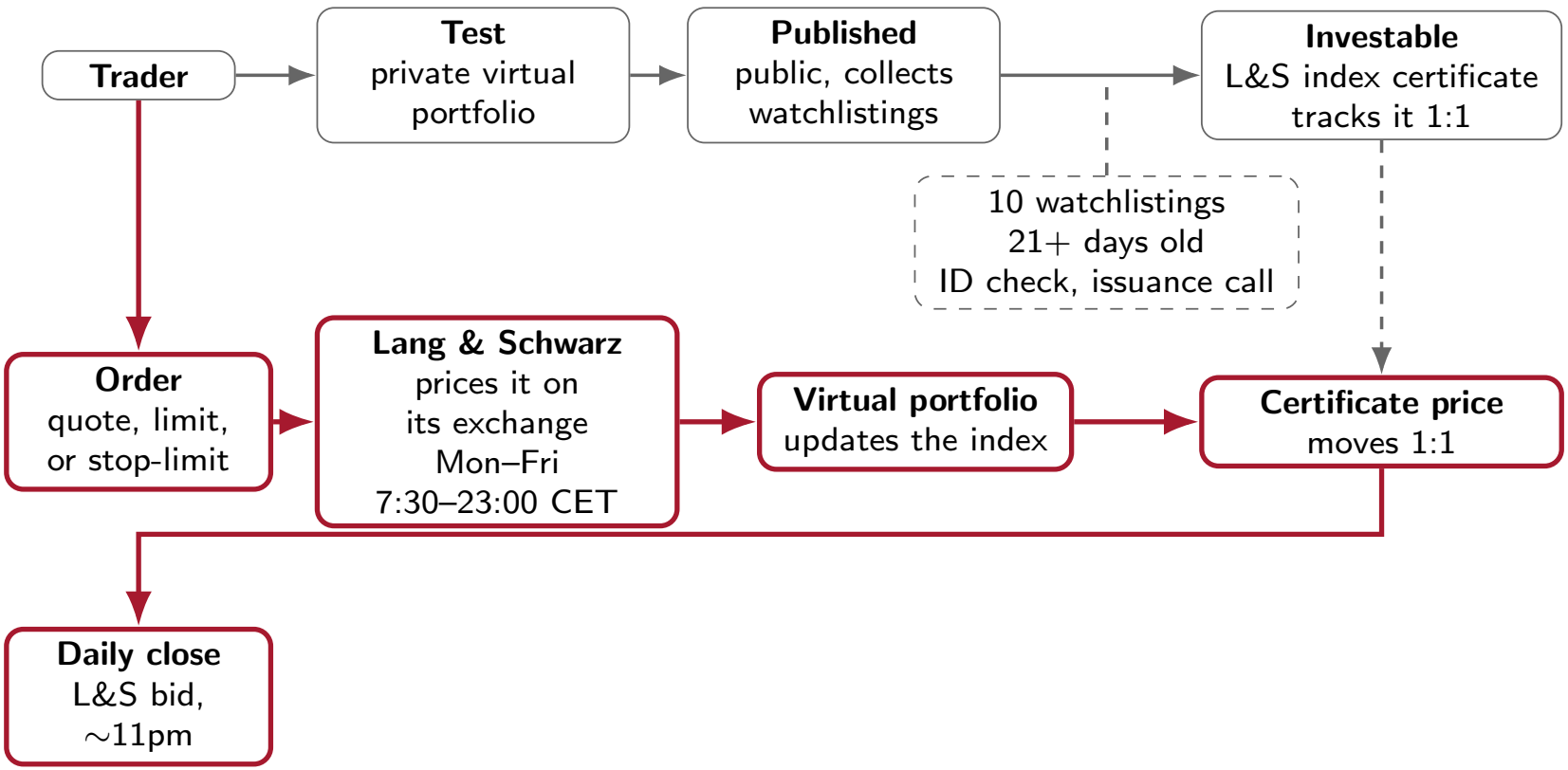
How a Wikifolio Works



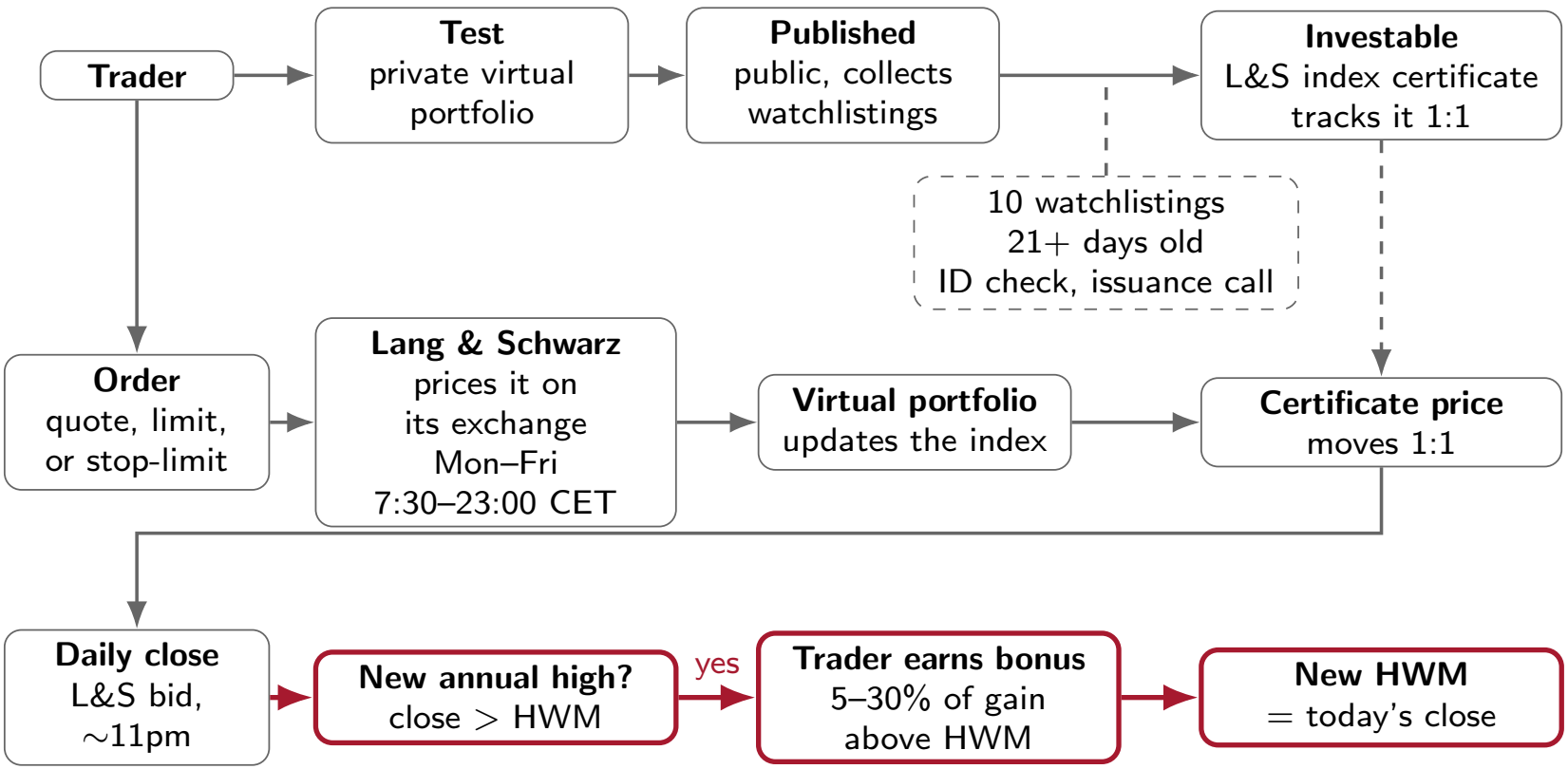
Trading

- Trades are virtual, priced at real-time quotes for the underlying asset on the exchange run by Lang & Schwarz
  - Quote orders (price valid for 3 seconds), limit orders, and stop-limit orders
- Trading hours (CET): Mon–Fri 7:30am–11pm, Sat 10am–1pm, Sun 5–7pm
- No transaction costs for trades inside the wikifolio
- With invested capital, Lang & Schwarz hedges the positions
- Daily close = Lang & Schwarz bid price at about 11pm

How a Wikifolio Works



How a Wikifolio Works



Sample High Water Mark

Date	Close Price	High Water Mark	Bonus Earned?
6/3/22	154	156	No
6/6/22	153	156	No
6/7/22	157	156	Yes
6/8/22	156	157	No
6/9/22	151	157	No
6/10/22	153	157	No
6/13/22	158	157	Yes
6/14/22	161	158	Yes
6/15/22	156	161	No
6/16/22	157	161	No

Bonus = (Close − HWM) × Rate; rate of 5–30% set by the trader; HWM resets January 1st

Data

	Current	**Extension**
Period	2018–2022	Through 2026
Underlying returns	Daily returns from CRSP and LSEG	Tick-level exchange and OTC tape data, quotes and trades (finalizing collection)

- **Filters:** drop closes below \$1, each fund's first month, January, and days over \$100 from a bonus; keep funds with 20+ trades a year and 6+ active months
- **Current sample:** 5,147 funds and 3.6 million fund-days (from 8,155 funds)

Portfolio Characteristics

	Mean	Median	Standard Deviation
Daily Trades	1.17	0	4.94
Daily Buys	0.66	0	3.02
Daily Sales	0.51	0	2.49
Daily Trades when Trades $>$ 0	5.43	3	9.48
Daily Quantity when Trades $>$ 0	5,206	100	349,310
Daily Dollar Amount when Trades $>$ 0	\$24,150	\$2,086	\$436,231
Portfolio Life (Years)	2.65	2.37	1.49
Daily Returns	0.02%	0%	2.05%
Daily Alpha	-0.0002%	-0.0001%	0.37%
Performance Fee	10.78%	10%	6.29%
Bonuses Earned Per Year	30.1	20	32.6

About 1 trade a day on average, and 5 on days with a trade

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67% of funds exhibit a negative alpha

Portfolio Characteristics

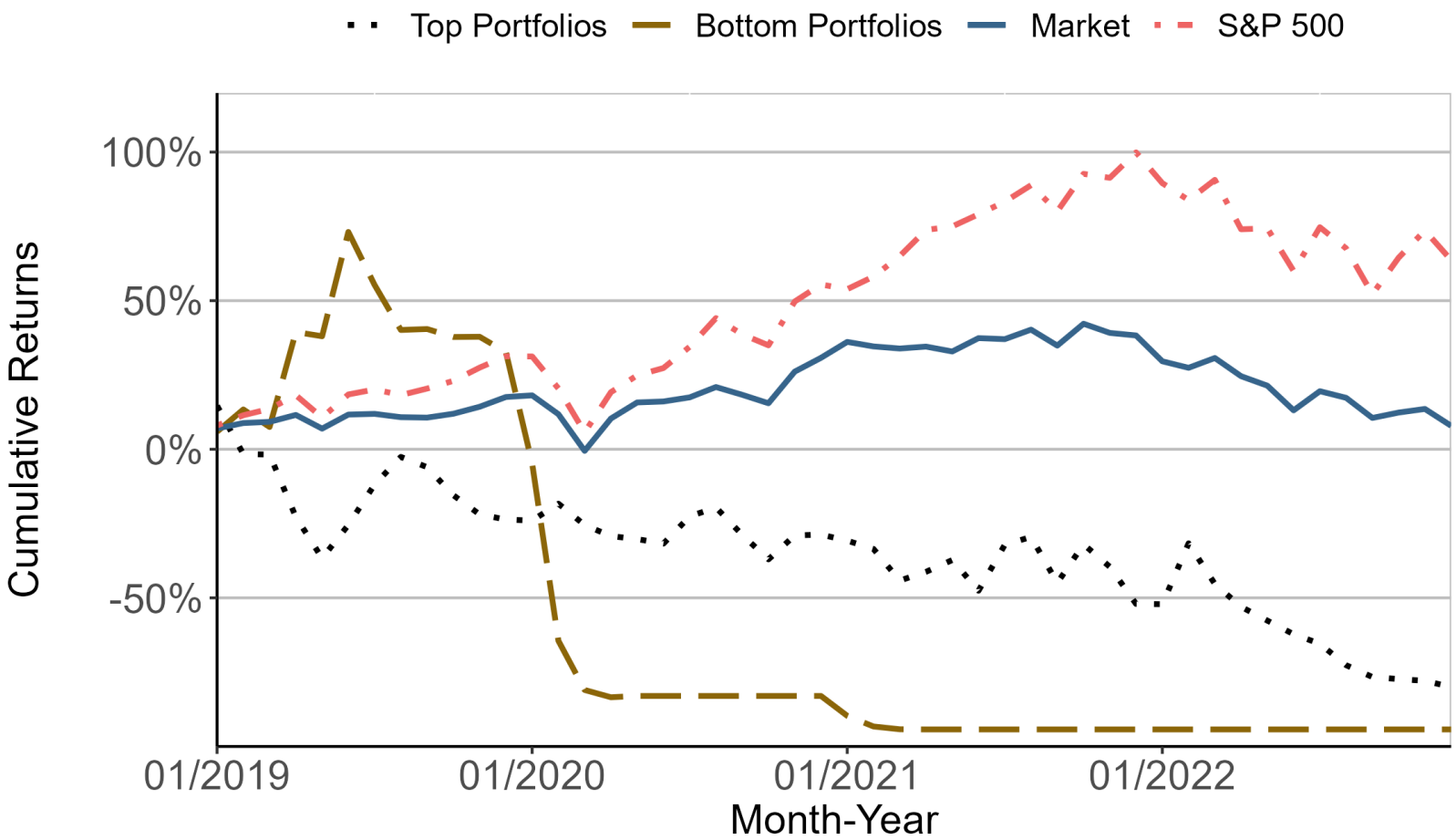
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About 30 bonuses a year on average (median 20)

Return Benchmarking

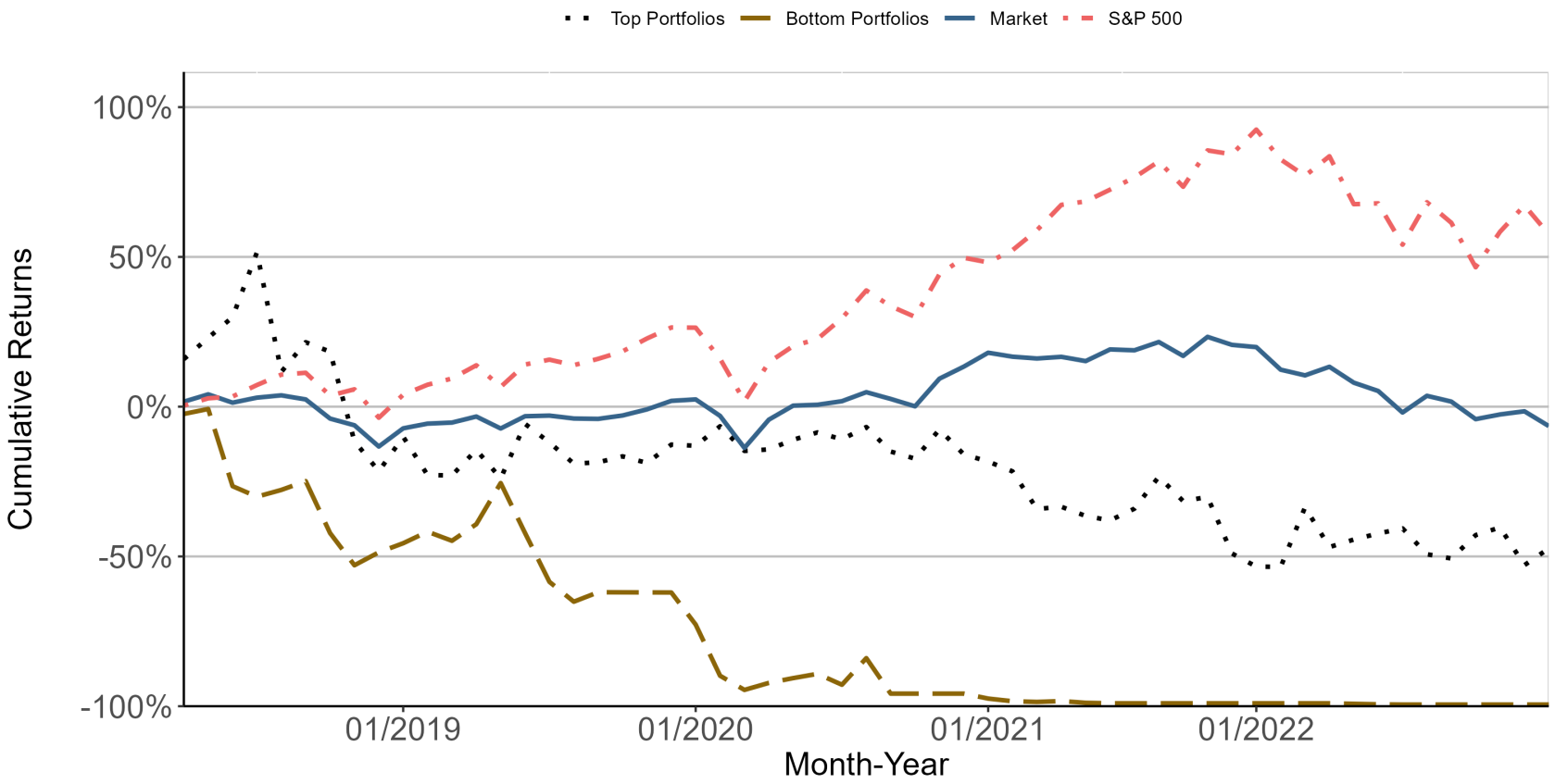
- Exercise
  - Rank funds by average return over the prior period
  - Invest in the ranked funds and hold for one period
  - Re-rank and rebalance either yearly, quarterly, or monthly
- Portfolios compared
  - **Top Portfolios:** 10 funds with the highest prior-period return
  - **Bottom Portfolios:** 10 funds with the lowest prior-period return
  - **Market:** equal-weighted, every fund active that period
  - **S&P 500:** benchmark, since 68% of funds trade US assets

Yearly Rebalancing



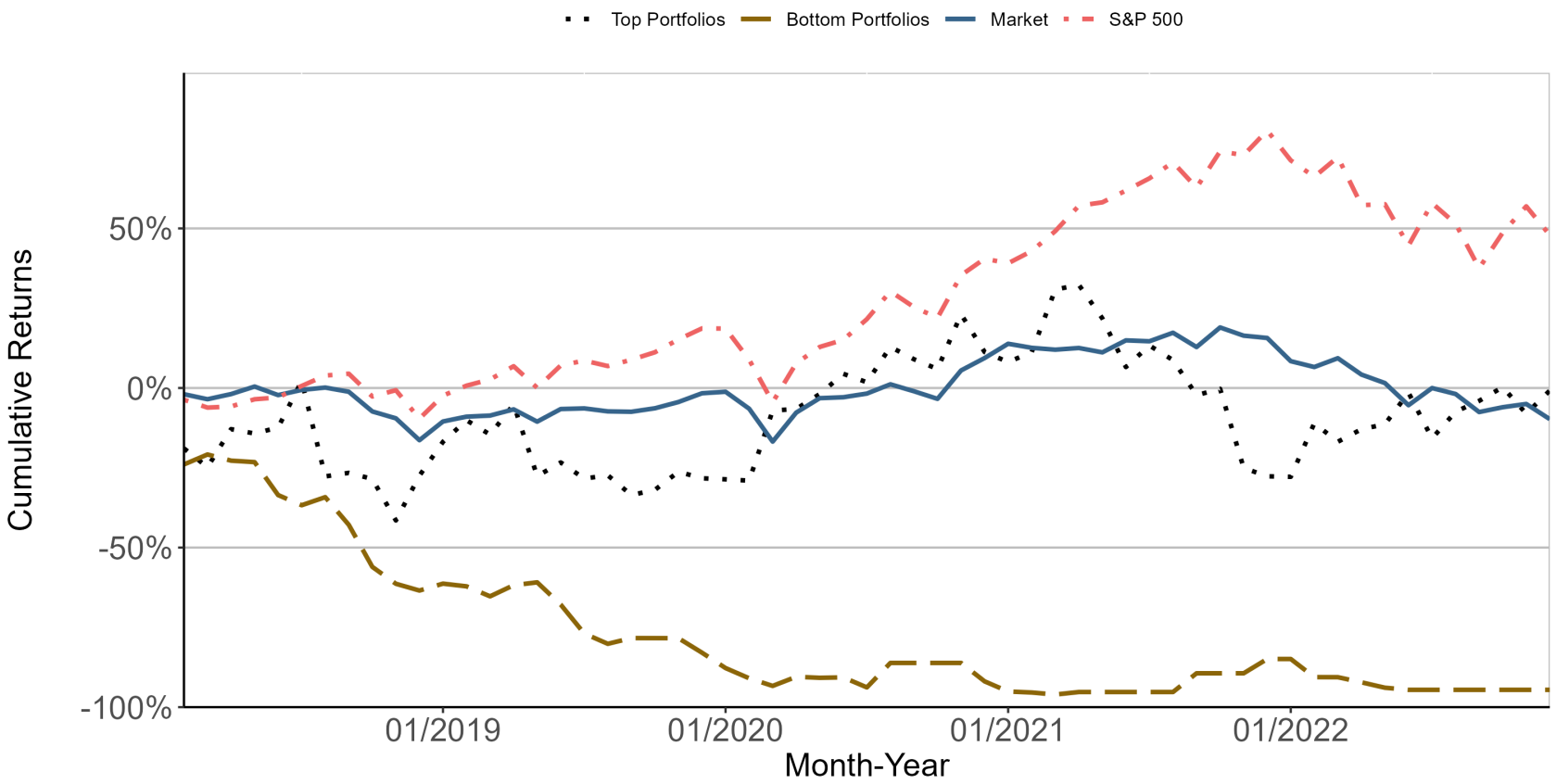
Both the top and bottom funds yield almost complete losses

Quarterly Rebalancing



Slight improvement for the top funds, which only lose about 50%

Monthly Rebalancing

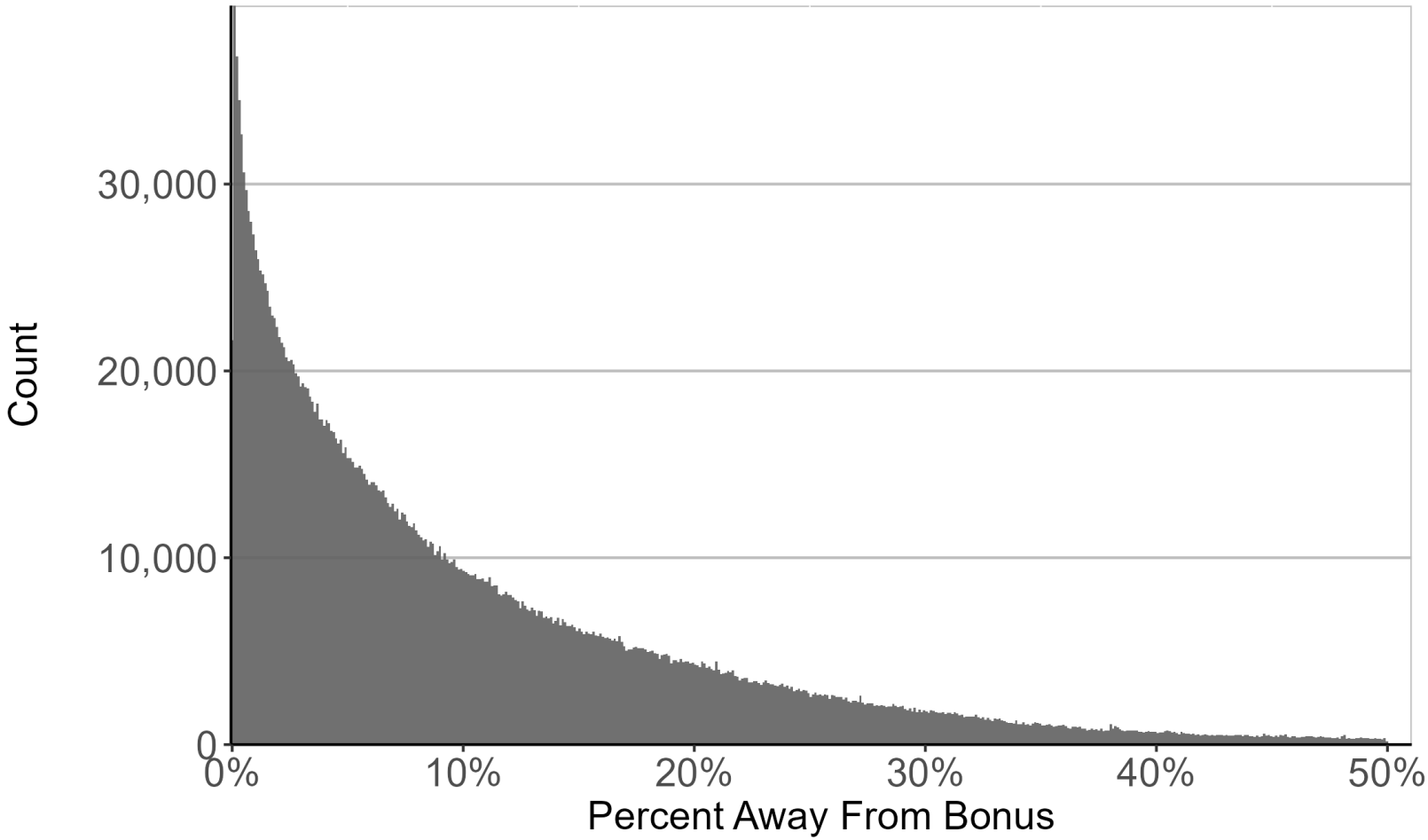


Top performers break even, but no consistent performance

Why Doesn't Performance Persist?

- In line with traders favoring short-term strategies, rather than buy and hold
  - Most traded: leveraged ETFs on market indices, and stocks like Apple, Tesla, Amazon, and Nvidia
- Daily HWM could encourage this
- **How close are traders to a bonus on a typical day?**

Distribution of Distance to Bonus

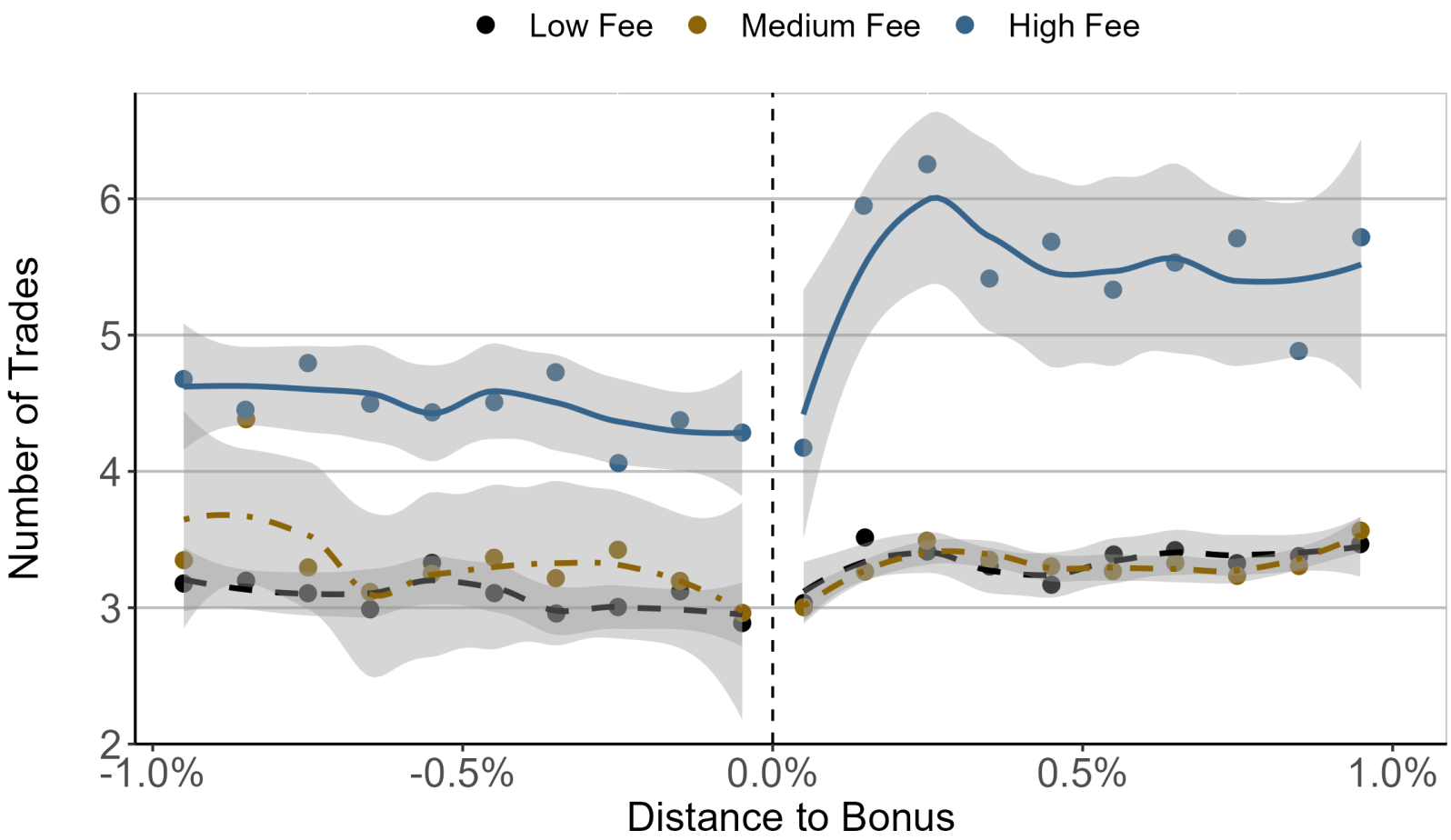


39% of all trader-days are within 5% of the bonus threshold

Is the Bonus Salient?

- Regression discontinuity around the bonus threshold
- Compare trader-days just above vs. just below the threshold (within $\pm 1\%$)
- Outcomes over the prior 3 trading days (traders average about 1 trade a day):
  - Number of trades
  - Whether any trade was made
- Split by bonus rate (fixed at creation):
  - Low: $\leq 7.5\%$ (1,881 traders)
  - Medium: 7.5–15% (2,375 traders)
  - High: $> 15\%$ (891 traders)

RDD Around Bonus By Group



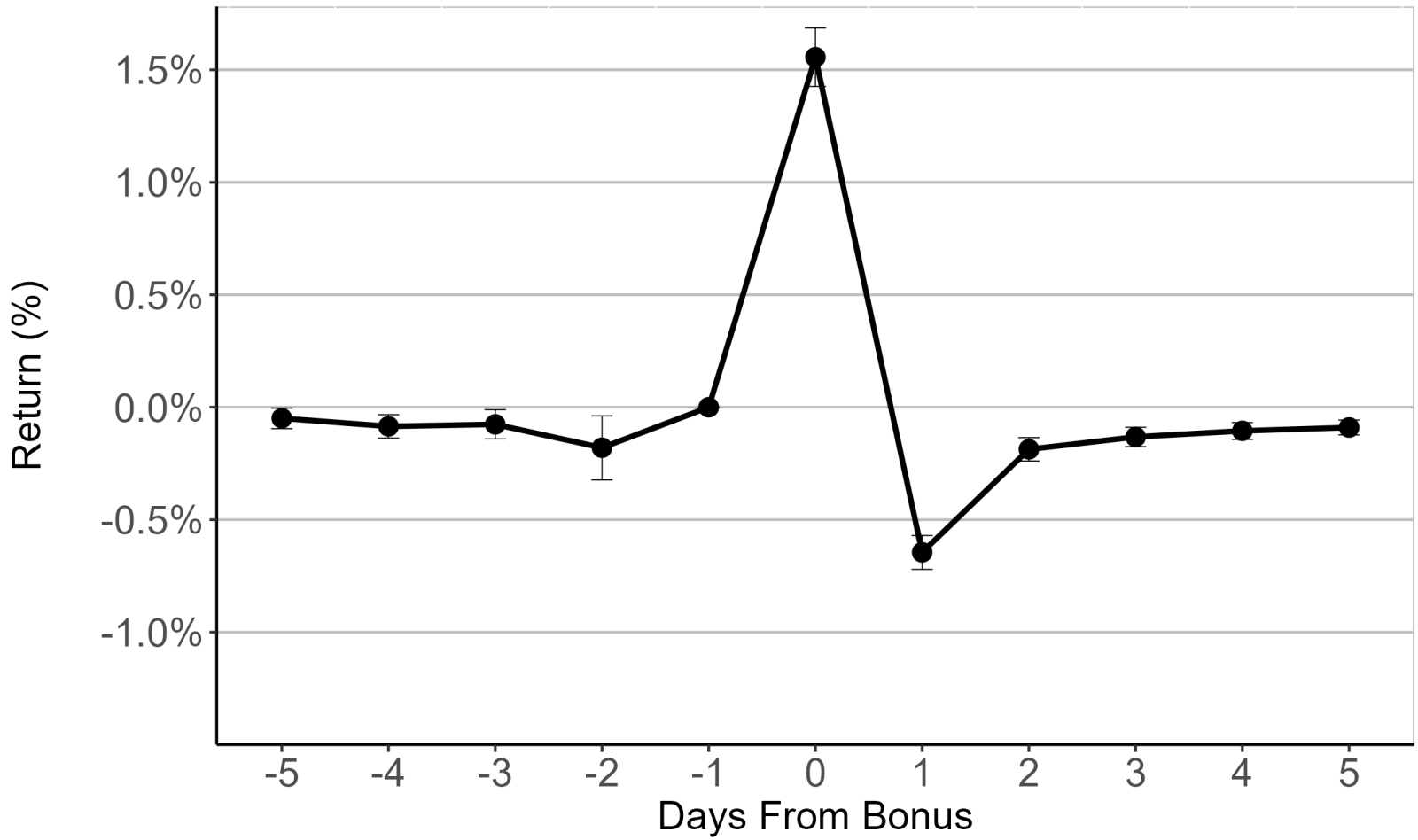
Higher bonus motivates greater trading around the HWM

Differences-in-Differences Estimator

- Treatment = earning a bonus, which is
  - **Non-absorbing:** the HWM resets every January 1st
  - **Non-binary:** traders earn many bonuses a year
- Two-way fixed effects is biased with heterogeneous, staggered effects (Callaway and Sant'Anna 2021; Sun and Abraham 2021)
- Estimator of de Chaisemartin and D'Haultfoeuille (2026): change from the day before a bonus, vs. funds with the same treatment history not yet treated

$$DID_{g,\ell} = Y_{g,F_{g-1}+\ell} - Y_{g,F_{g-1}} - \frac{1}{N_{g,F_{g-1}+\ell}} \sum_{g':D_{g',1}=D_{g,1},F_{g'}>F_{g-1}+\ell} \left[Y_{g',F_{g-1}+\ell} - Y_{g',F_{g-1}}\right]$$

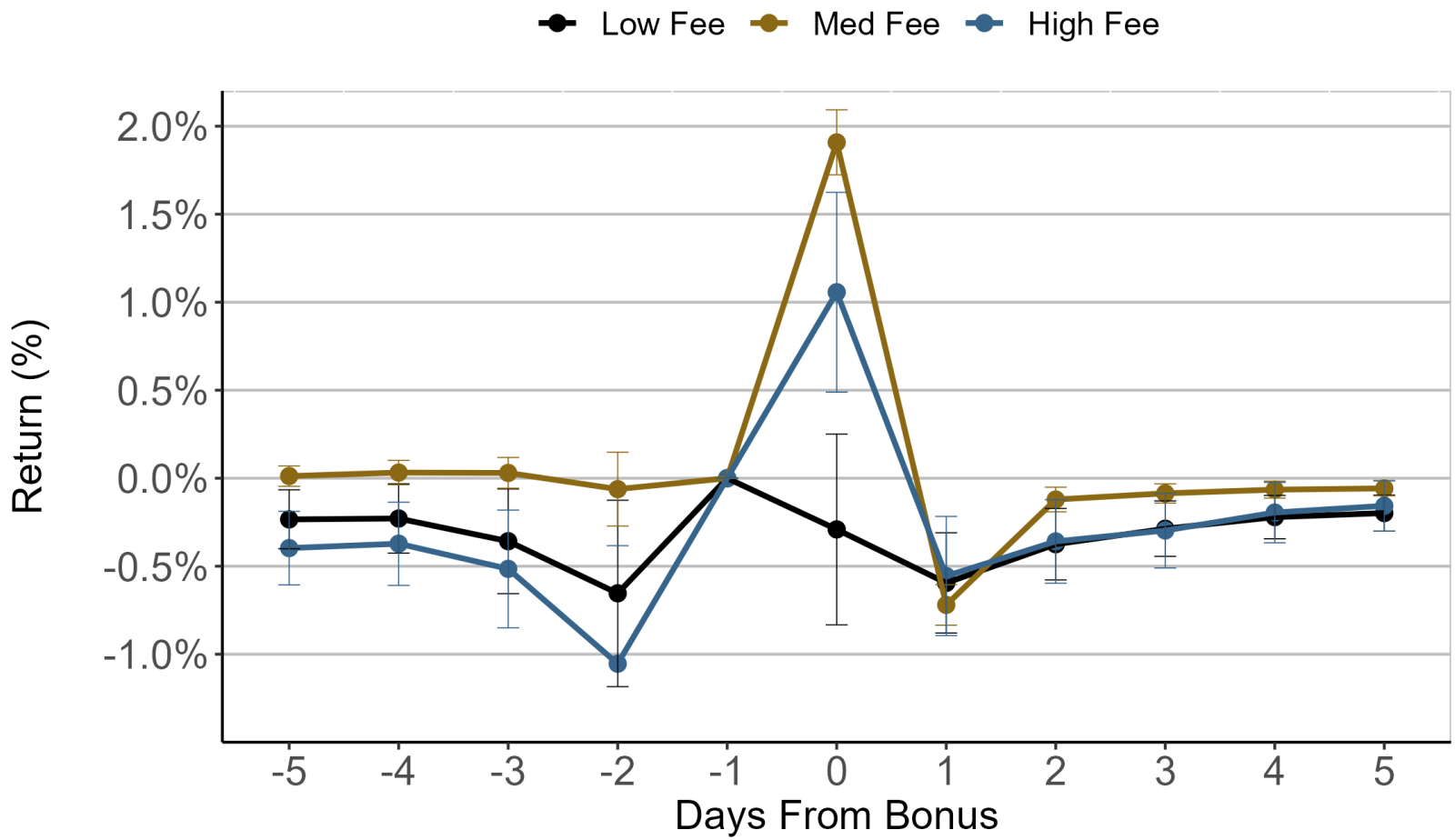
Returns Around Bonuses



Traders earn a 1.6 pp higher return on the bonus day

▸ Regression Table

Returns Around Bonuses By Performance Fee



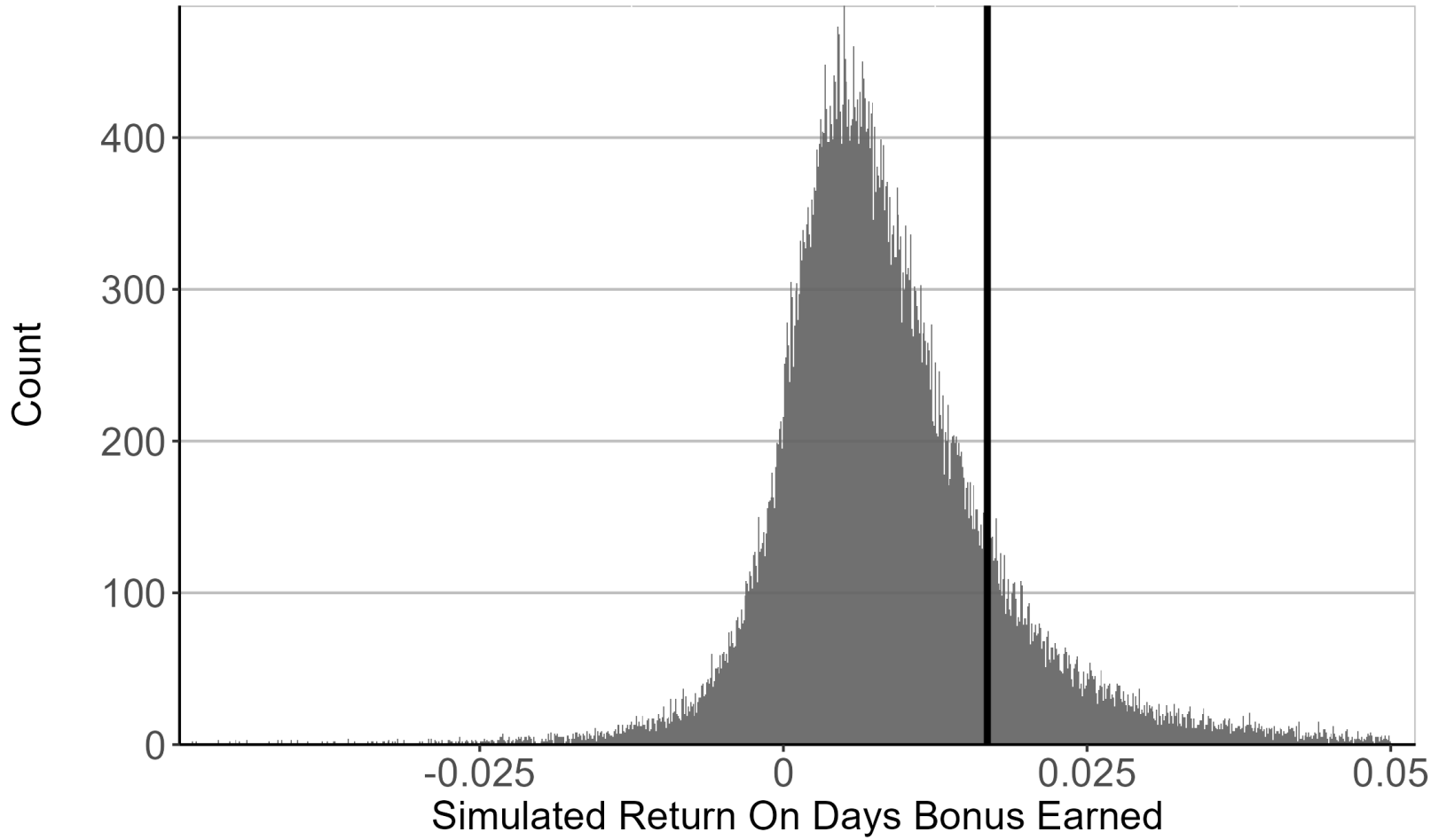
Higher fee traders have higher returns, consistent with incentives

▸ Number of Trades

Luck or Strategy?

- Returns must be positive on a bonus day, so how big is a 1.6 pp jump?
- Luck vs. skill bootstrap (Kosowski et al. 2006; Fama and French 2010)
- For each bonus day:
  - Keep the fund's portfolio weights from the prior day
  - Draw that day's asset returns at random from the assets held by all funds crossing that day
  - Repeat 10,000 times
- Compare the simulated returns to the actual bonus-day return

Bootstrap Return Distribution

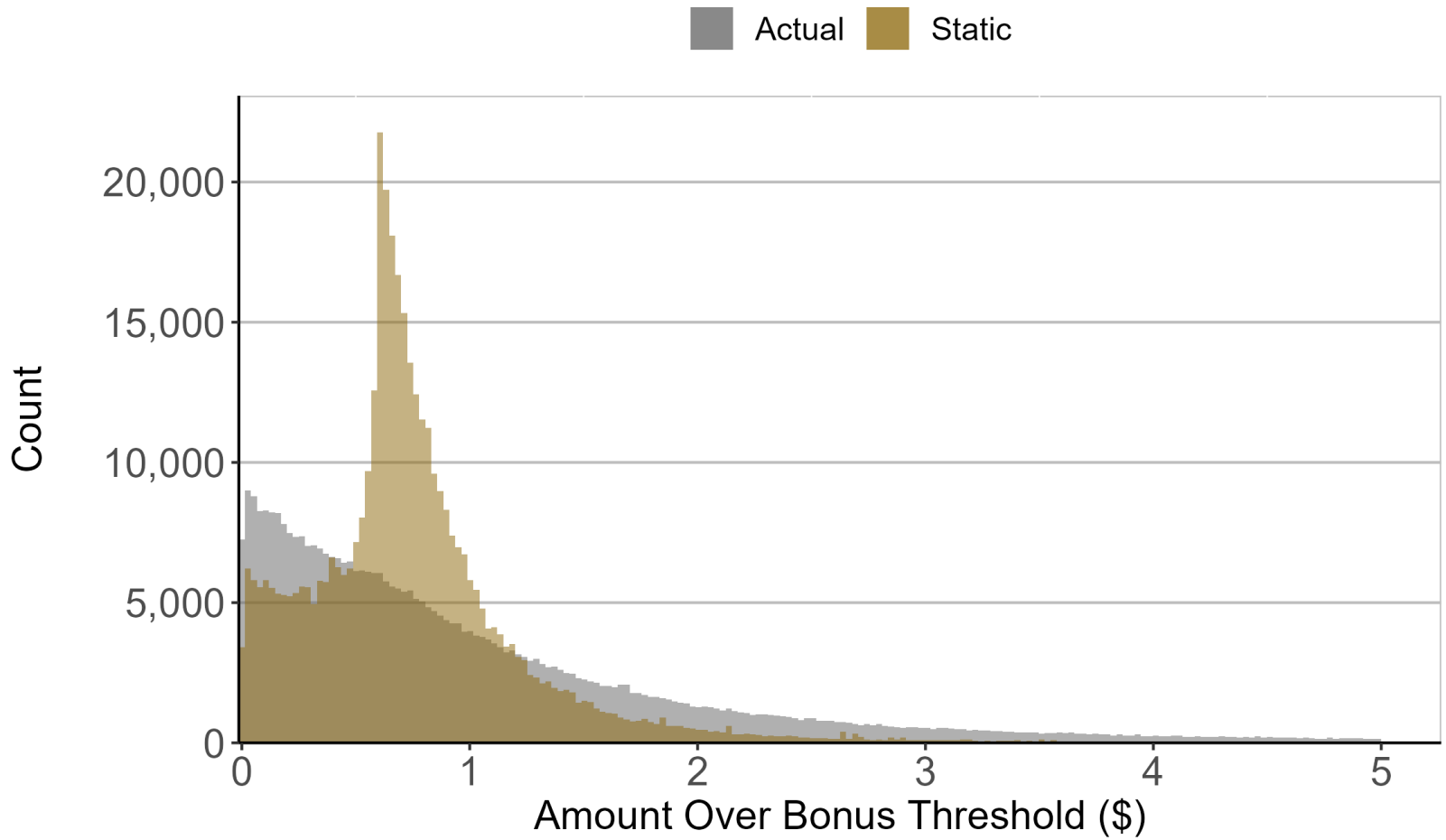


Observed 1.6% return is larger than 82% of simulations

Static Portfolio Counterfactual

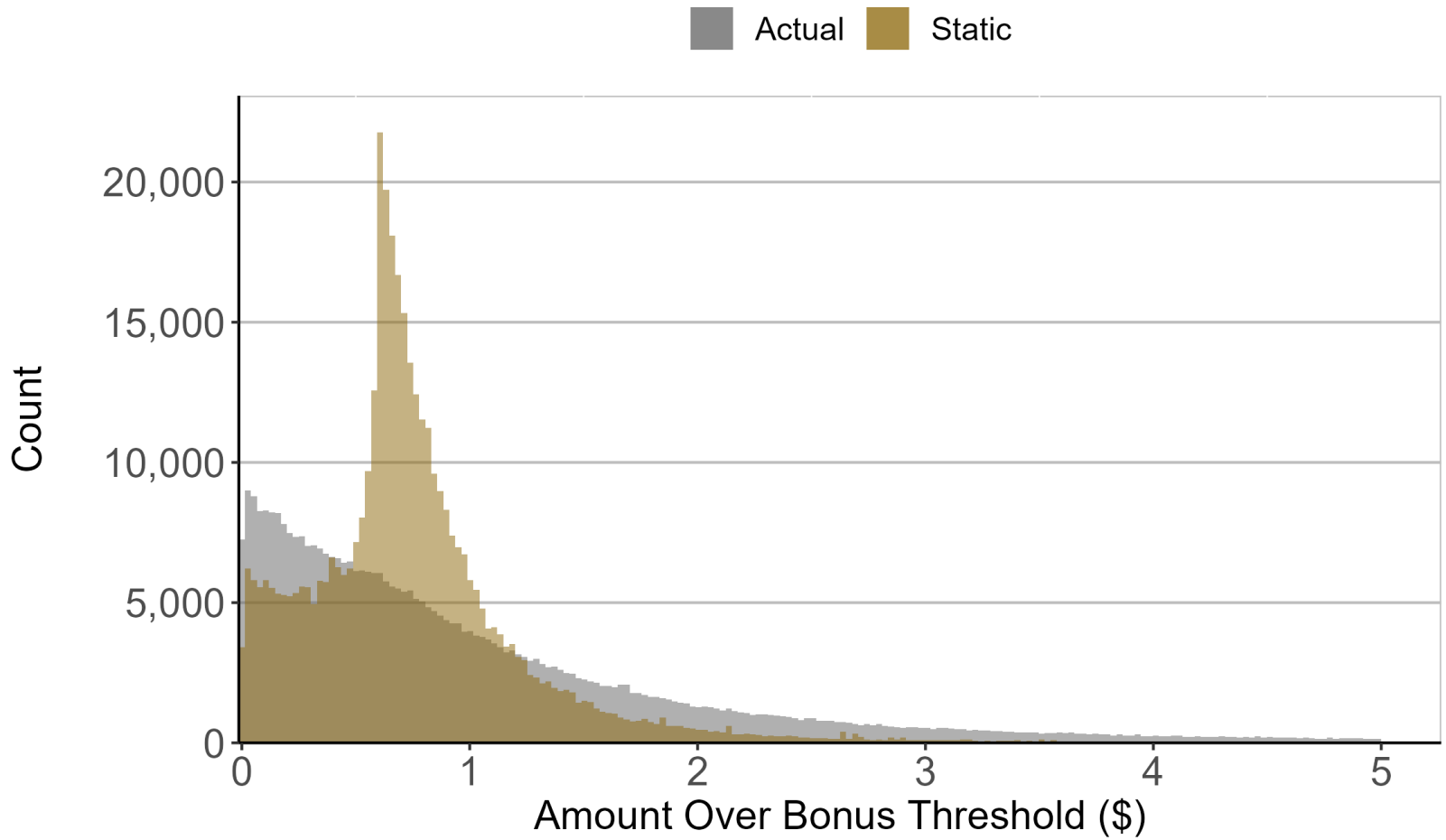
- What if traders did not trade at all?
- Freeze each fund's holdings on February 1st and hold them for a year
- Cumulative returns are similar and FF3 alpha is higher under a static setting for 63% of funds
- But is this optimal for bonus achievement?

Distance Over Bonus Threshold



Under the static trading strategy 57% of fund-years achieve more bonuses

Distance Over Bonus Threshold



Long right tail of actual crossings, consistent with HWM incentive

Summary of Additional Results

- **Mechanism** ▸ Details
  - Assets traded on bonus days had higher past 10-day returns (sales 7.4% vs. 4.8% for purchases)
  - Sold assets went on to earn more: 0.69 pp higher future returns than purchases
  - Sales jump from about 40% to 48% of trades on bonus days
- **Investors (new data, preliminary)** ▸ Details
  - Certificate trades rise about 25% on bonus days (top 20% most-traded)
  - Volume rises too, but the estimate is noisy
- **January reset** ▸ Details
  - All HWMs reset on January 1st, so bonus timing comes from the calendar
  - Similar return jump on bonus days as the full sample

Conclusion

- Spike in returns corresponds with selling high-return assets and forgoing potential further gains
- High-frequency incentives can be counterproductive for retail investors
  - Exacerbating short-term biases and undermining sustainable performance
- Highlights importance of incentive frequency and interaction with agent type

Thanks!

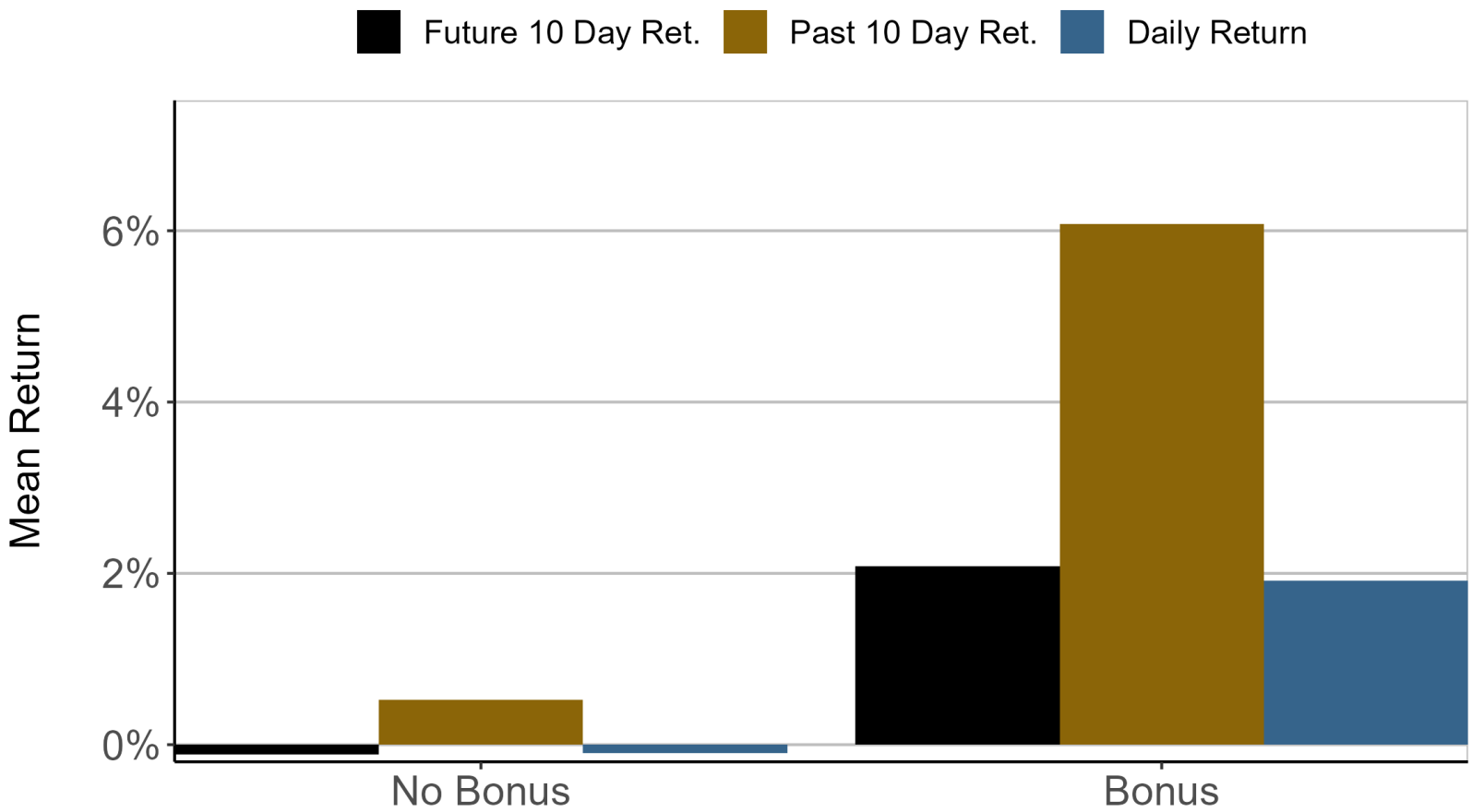
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Mechanism: What Do Traders Trade?

- Use transaction-level data
- Compare assets bought and sold on bonus vs. non-bonus days
- Look at each asset's past 10-day, same-day, and next 10-day return

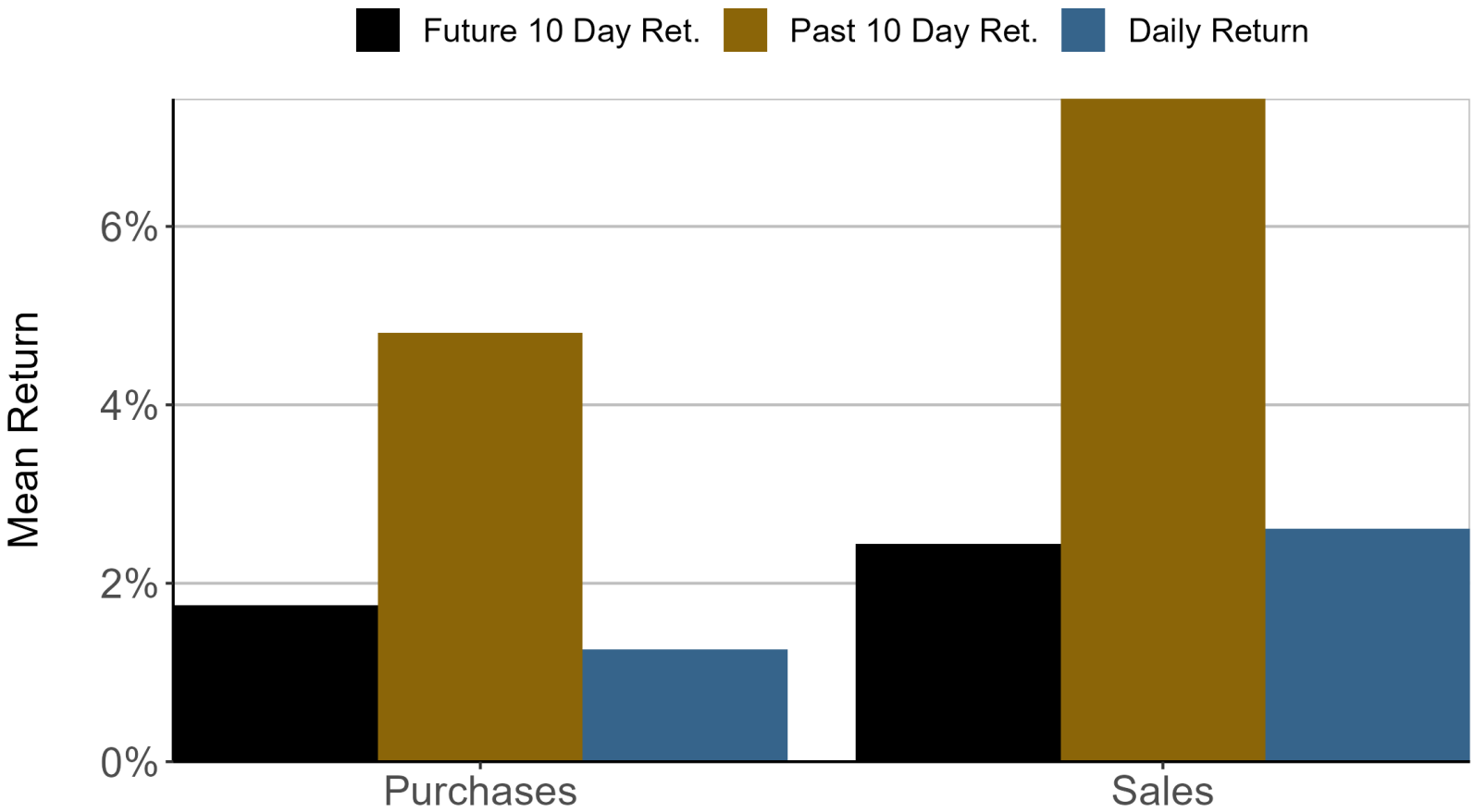
Return Patterns: Bonus vs. Non-Bonus Days



Assets traded on bonus days had higher past returns

▸ Sales Table ▸ Purchases Table

Return Patterns: Purchases vs. Sales on Bonus Days



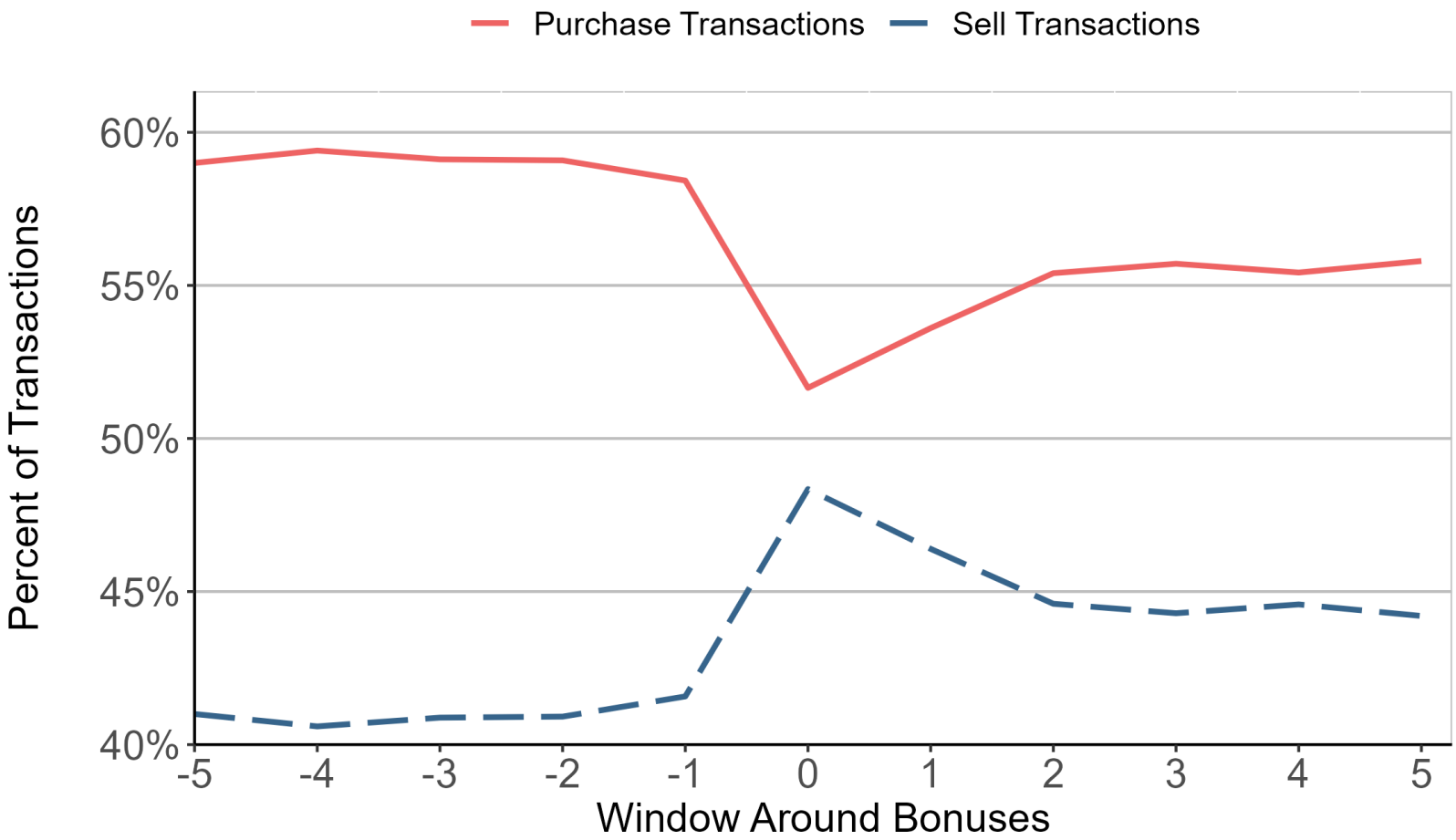
Future returns are larger for assets that traders sold on bonus days

▸ Sales Table ▸ Purchases Table

Selling to Lock In the Bonus

- Are traders selling recent winners to secure the bonus?
- Share of transactions that are sales, in the 10-day window around a bonus

Buying vs. Selling Activity



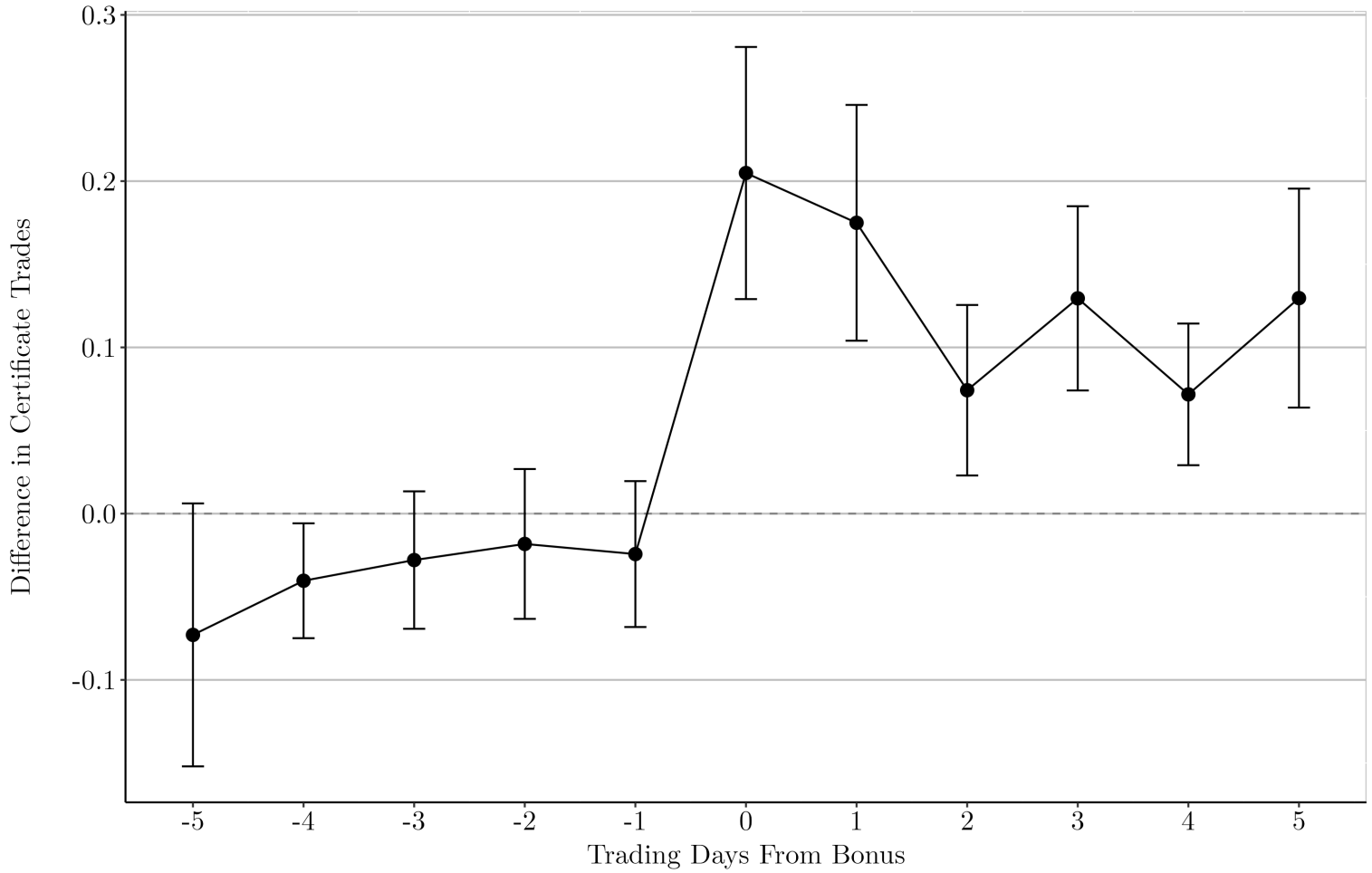
Consistent with selling recent winners to lock in the close price

Back to Summary

New Data: How Investors Trade the Certificates

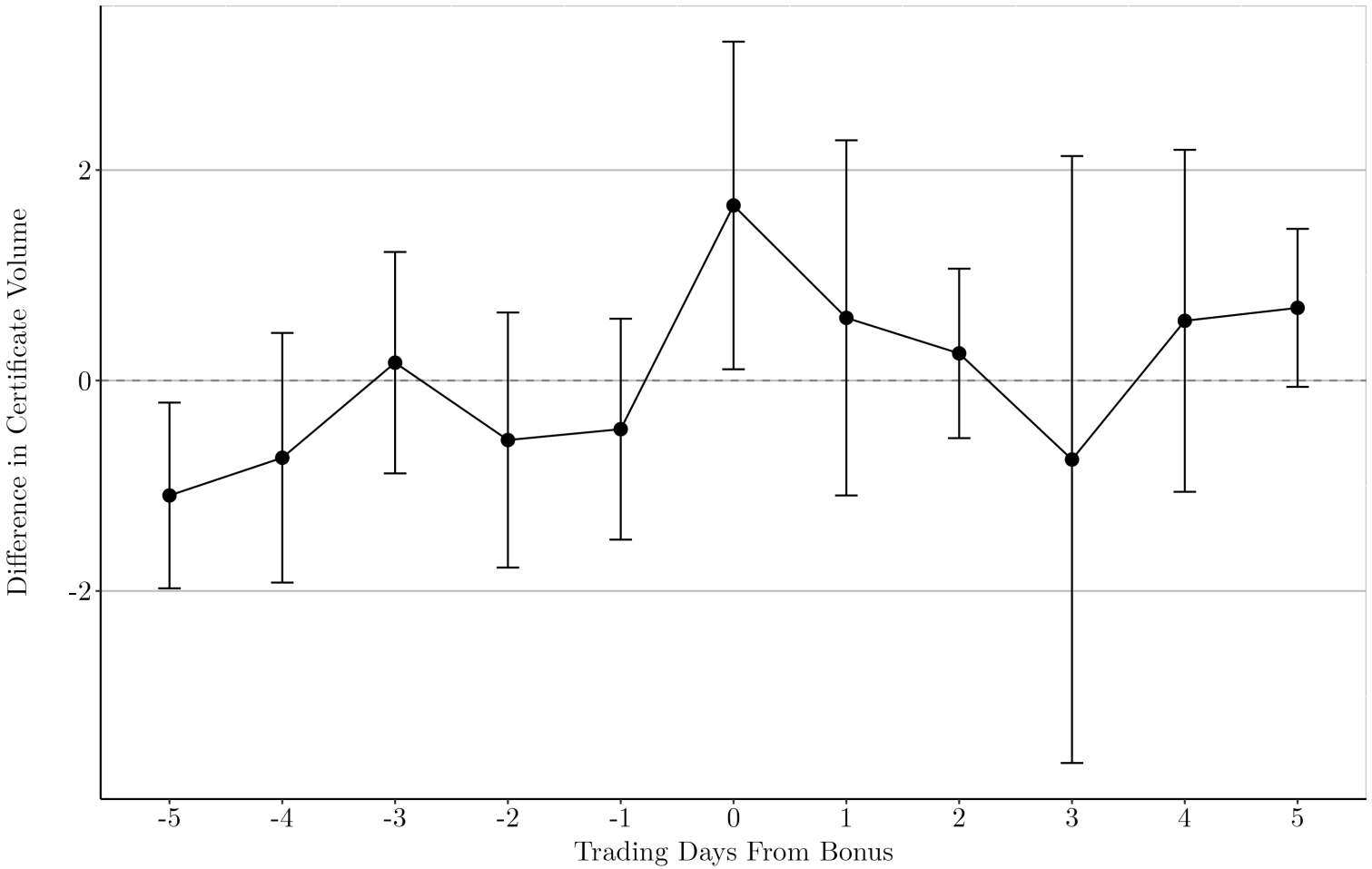
- Newly collected certificate trading data (preliminary results)
- Do investors respond to bonus days?
- Event study of certificate trades and volume around bonuses, in the most-traded certificates

Certificate Trades Around Bonuses: Top 20% Most-Traded



∼25% of the 0.82 daily mean for these certificates

Certificate Volume Around Bonuses



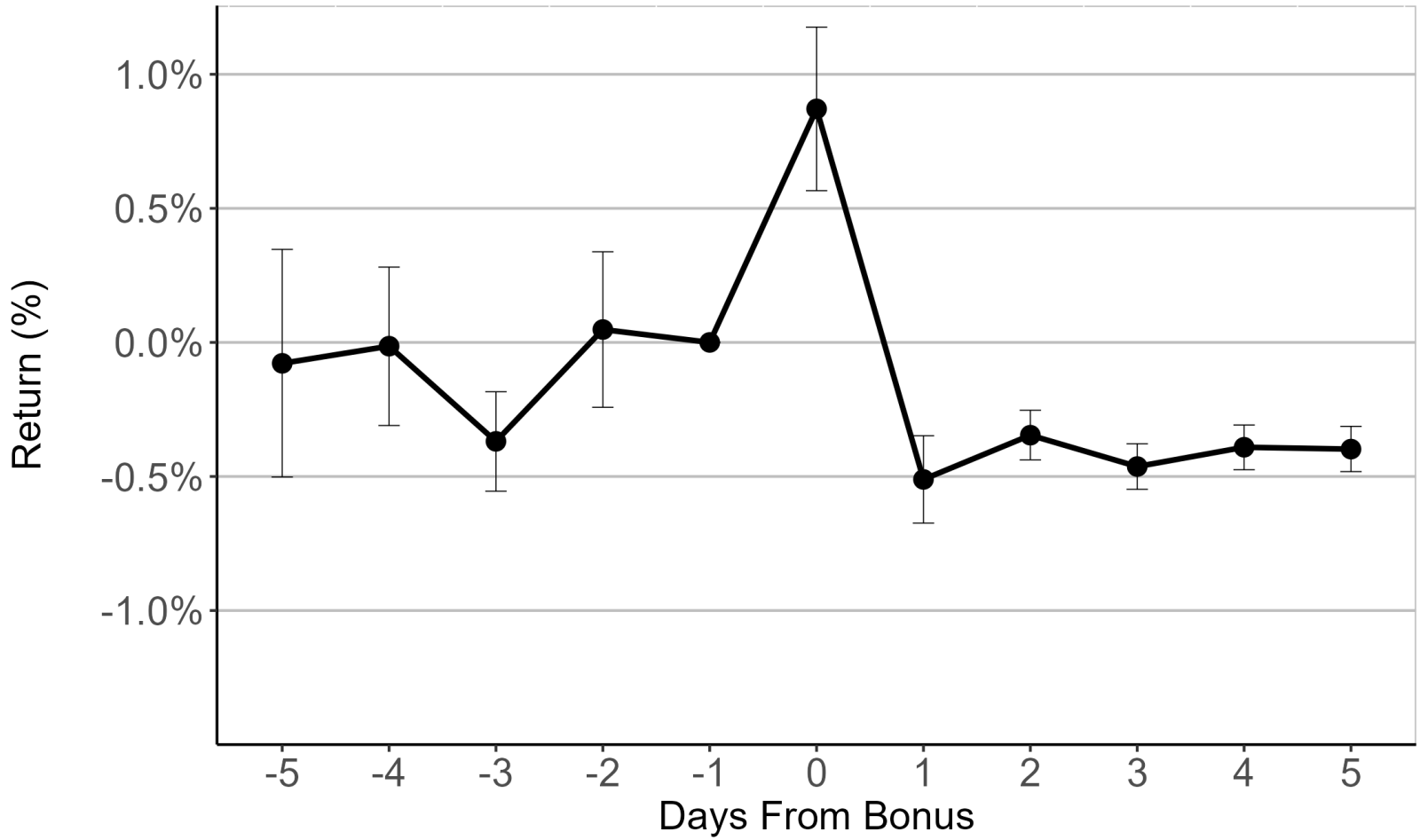
∼23% of the 7.29 daily mean for these certificates

Back to Summary

January Reset

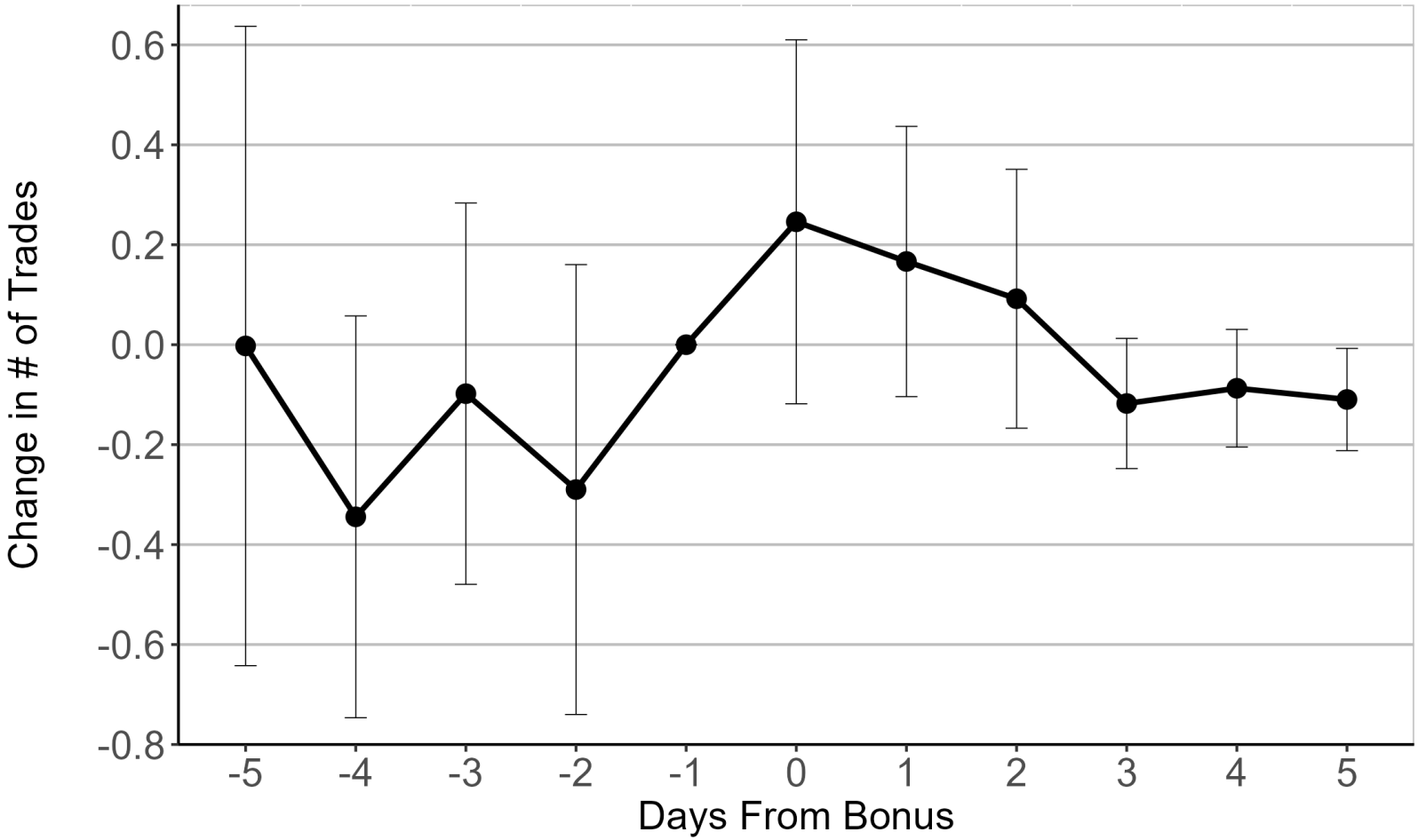
- Concern: more skilled traders may cross earlier or more often, and crossing depends on past performance
- Every January 1st all HWMs reset to that day's close
  - More easily achievable bonus allows us to isolate its effect
- Re-run the DiD on January only, first 3 bonuses per trader (437,046 observations)

January: Returns Around Bonuses



Similar pattern to the whole sample

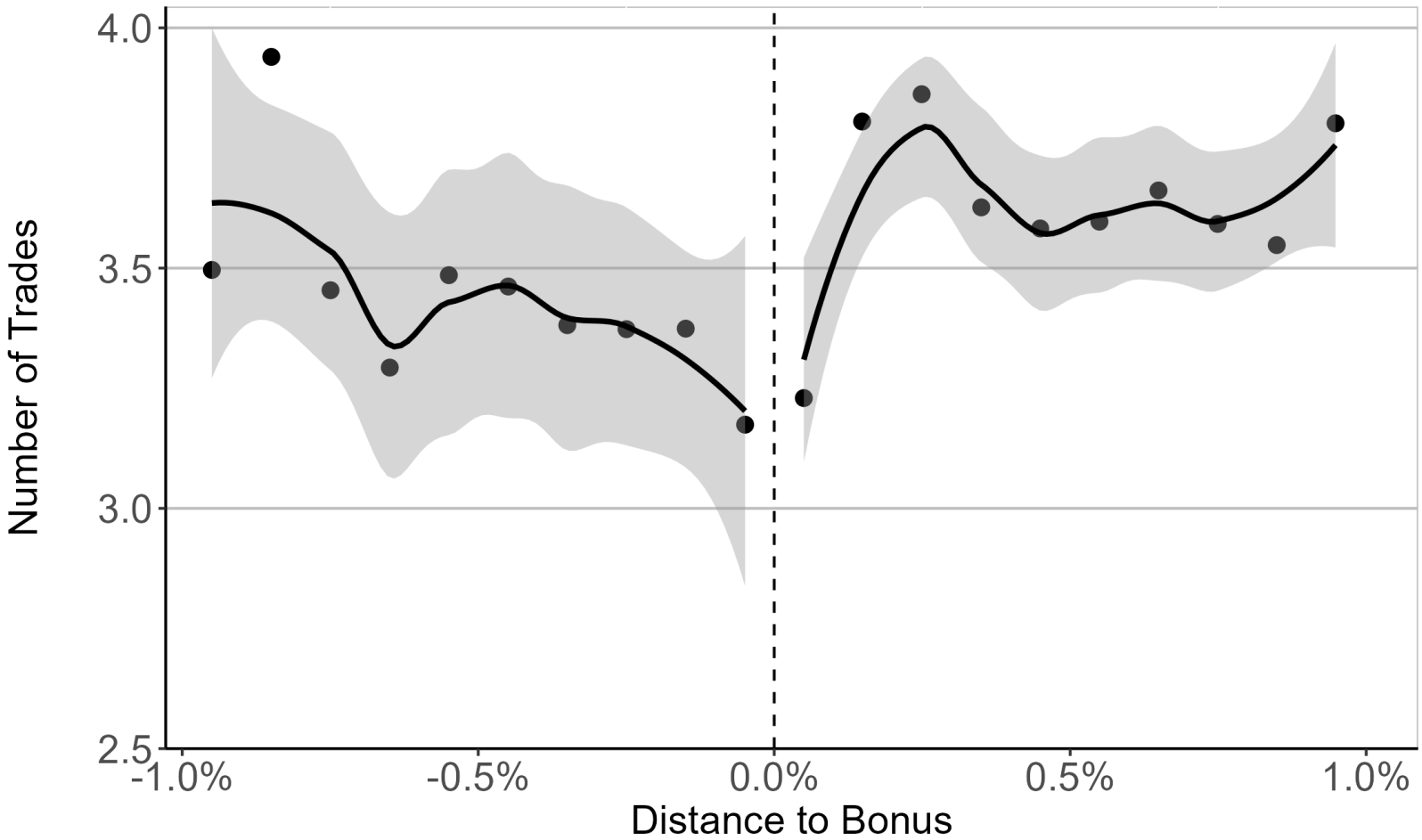
January: Number of Trades Around Bonuses



Noisy, yet positive, estimate of trades

Back to Summary

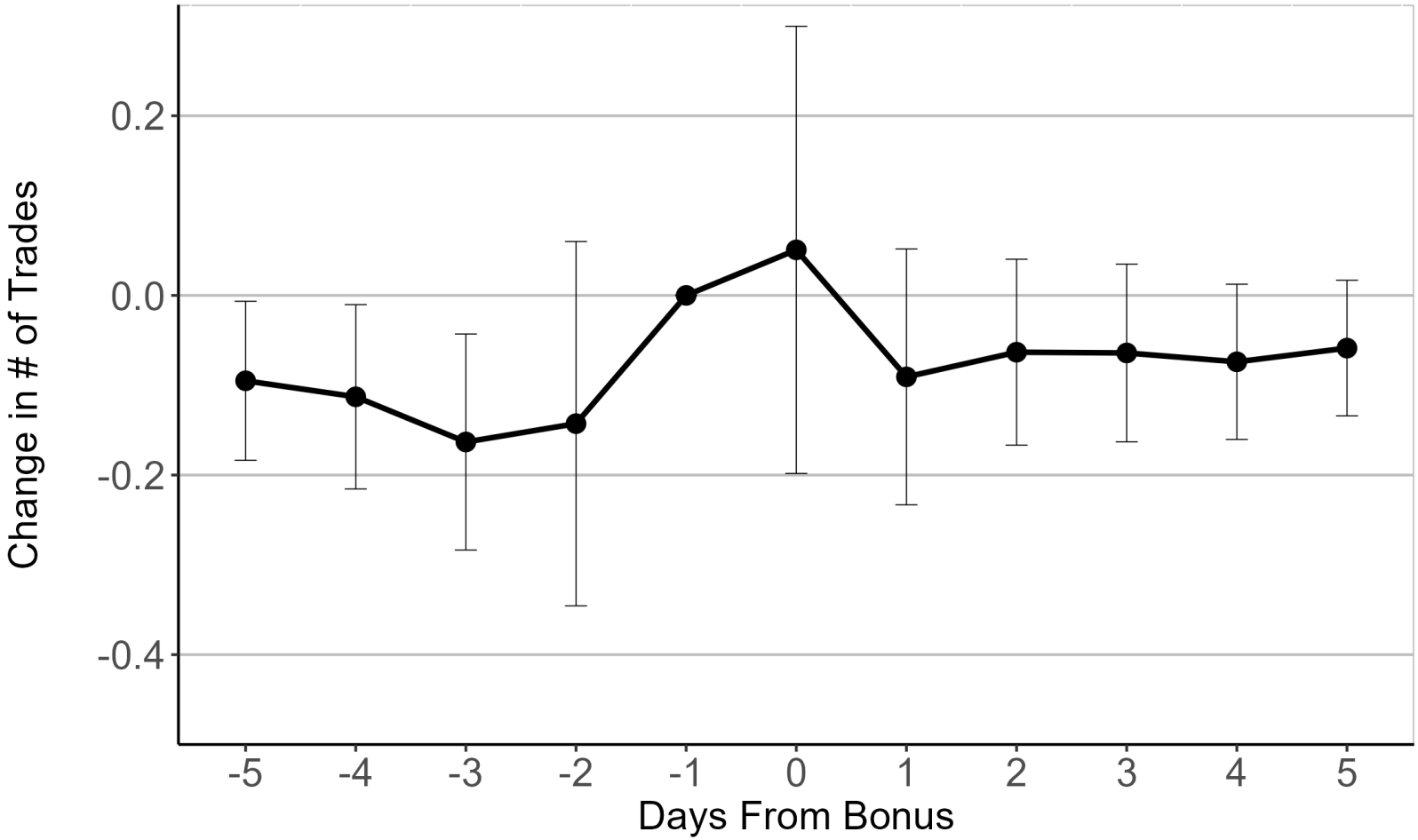
RDD Around Bonus



Intensive: 0.31 trades (9% of mean) | Extensive: odds increase 15%

▸ Regression Table

Number of Trades Around Bonuses



No significant increase in trades placed

▸ Regression Table

Who Pushes Hardest for the Bonus?

- If the bonus drives the return jump, traders with the most to gain should push hardest
- A higher bonus rate means a larger payout from crossing the HWM
- Test: re-run the DiD around bonuses separately for low, medium, and high fee traders
- Prediction: the bonus-day return jump shows up in the higher-fee groups

Back of the Envelope

- Is chasing the bonus worth it for the trader?
- **Gain:** the bonus as a share of fund value, $Bonus_{i,t}/Price_{i,t}$
- **Cost:** abnormal return, the next 10-day return after the bonus minus the 10-day return just before it

$$Net_{i,t} = \frac{Bonus_{i,t}}{Price_{i,t}} + \left(R_{i,t}^{+10} - R_{i,t}^{-10} \right)$$

- $Net_{i,t} < 0$: the push for the bonus cost more than it paid
- Average net loss of 2.5% of fund value; median of -1.4%
  - Statistically different from zero; robust to window length and to overlapping bonus windows

Social Trading (Wikifolios)

- Individuals register and build a virtual portfolio using over ∼600,000 financial products, including
  - Stocks
  - Exchange-traded products
  - Funds (real estate, bonds, equities, money market, or mixed funds)
  - Investment certificates (index certificates, bonus certificates, discount certificates, and reverse convertibles)
  - Leverage products
- Traders trade virtually, but prices reflect real-time bid/ask quotes
- An index is created to replicate the virtual portfolio's performance 1:1

Starting a Wikifolio

- Any registered user can create a wikifolio: a virtual portfolio plus a written trading idea (at least 200 characters)
- The trader chooses the investment universe the wikifolio can trade
- **Test** status: private, only the trader can see it
- To publish: trader profile, title, idea, at least one trade, and verified email
- **Published** status: searchable, users can add it to their watchlist and follow every trade

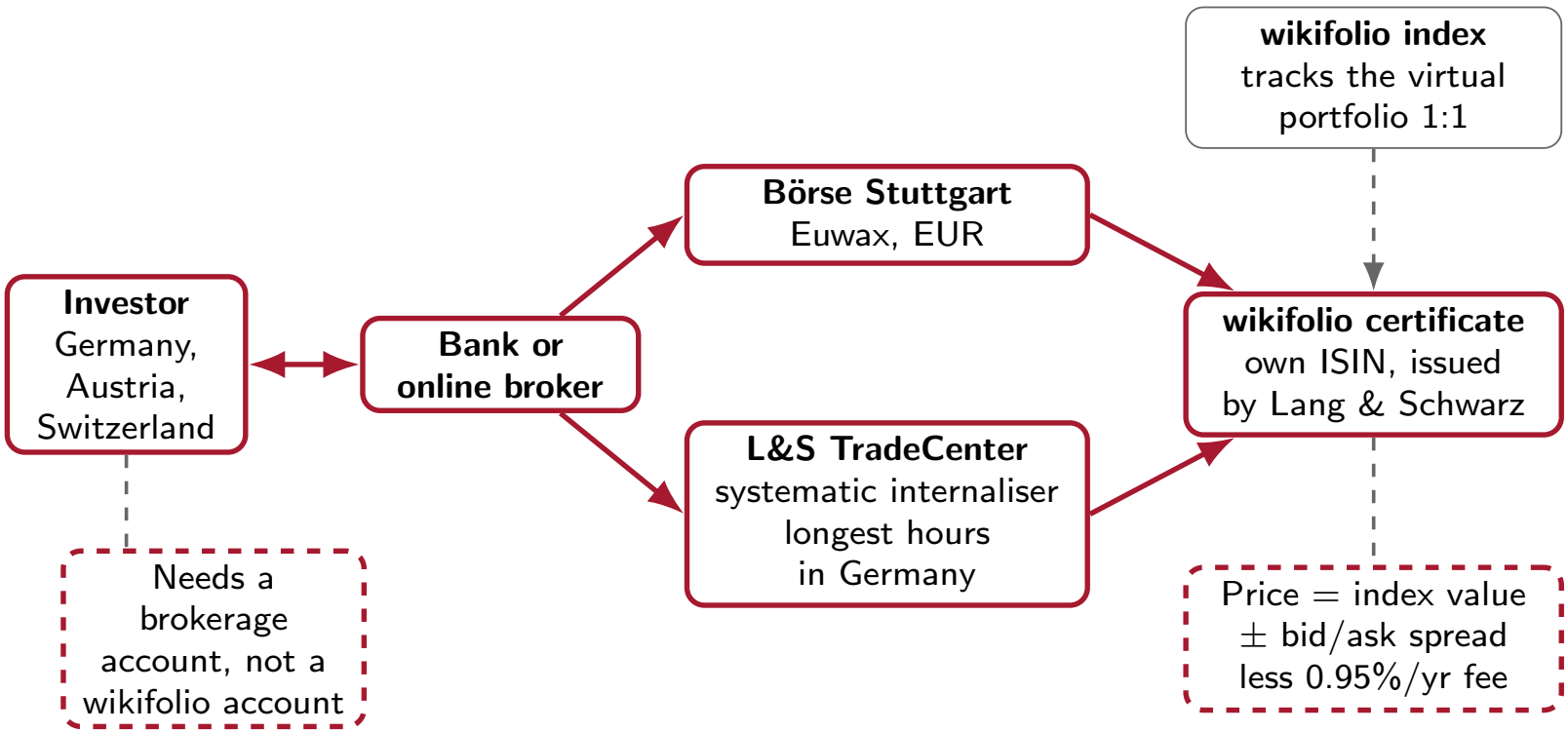
Becoming Investable

- At least 10 watchlistings and the wikifolio is at least 21 days old
- Legitimation: ID, address, and phone number (18 or older)
- Issuance call with wikifolio about the trading idea and strategy
- Lang & Schwarz issues an open-ended index certificate with its own ISIN that tracks the wikifolio 1:1
- Once issued, the investment universe is fixed

Investing in Wikifolio Certificates

- Investors buy the certificate by its ISIN through their own bank or online broker, with no wikifolio account needed
- The certificate is not traded on the Lang & Schwarz exchange, but on
  - Börse Stuttgart (Euwax, in EUR), or
  - Directly with Lang & Schwarz TradeCenter: a systematic internaliser, Germany's off-exchange venue with the longest trading hours
- Who can buy
  - Germany: every certificate is publicly offered
  - Austria: certificates of Austrian traders or with high demand
  - Switzerland: only stock-only wikifolios (BX Swiss, in CHF)
- Price tracks the wikifolio 1:1, with a bid/ask spread from its holdings and a 0.95% annual certificate fee charged daily

How Investors Buy In



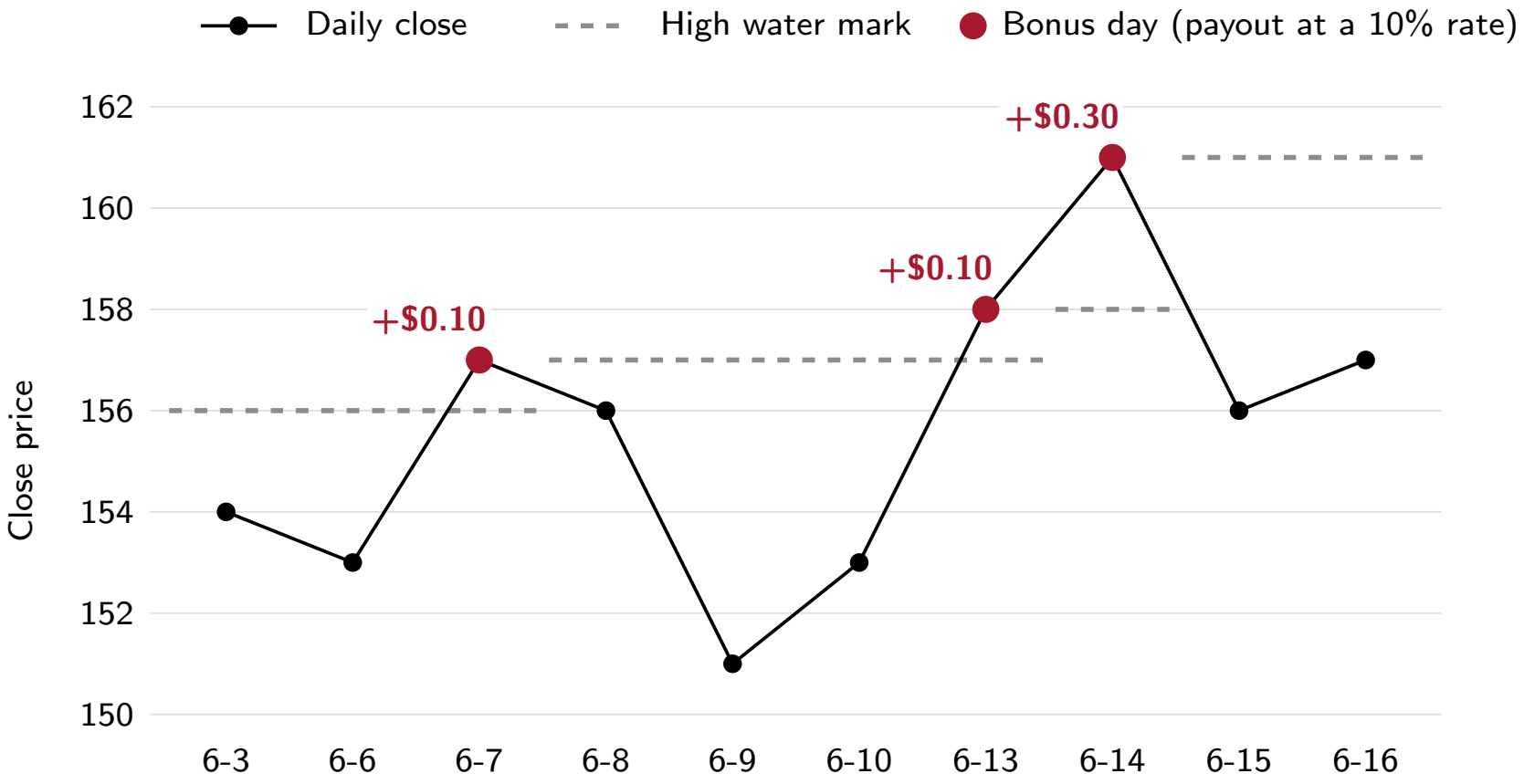
Fees

- Traders earn no fee on assets under management, only performance bonuses
- Bonus rate of 5–30% of the gain above the high water mark (HWM), set by the trader at launch and fixed
- Earned only when the daily close sets a new annual high
- The HWM resets every January 1st

Bonus Calculation

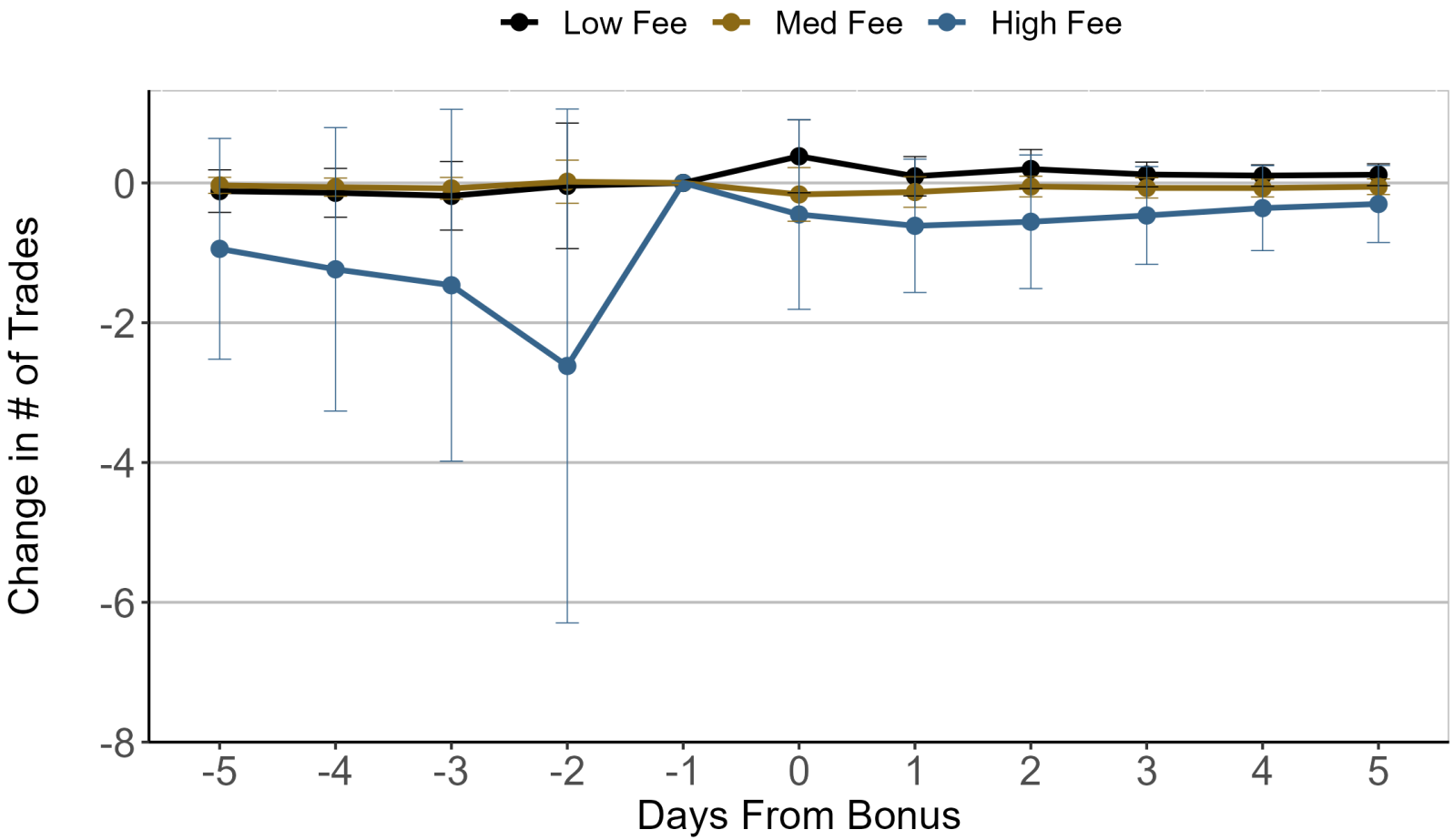
$$Bonus_{i,t} = \begin{cases} (Price_{i,t} - HWM_{i,t}) * Rate_i, & \text{if } Price_{i,t} > HWM_{i,t} \\ 0, & \text{otherwise} \end{cases}$$

Sample High Water Mark



Bonus = (Close − HWM) × Rate

Number of Trades Around Bonuses By Performance Fee



No difference across fee

Returns By Performance Fee

RDD Around Bonus Achievement

	Number of Trades	Probability of Trading
	(1)	(2)
Intercept	3.237***	-0.497***
	(0.037)	(0.007)
I(Distance to Bonus $\geq$ 0)	0.312***	0.140***
	(0.051)	(0.009)
Distance to Bonus	-41.158***	-17.497***
	(6.745)	(1.244)
I(Distance to Bonus $\geq$ 0)$\times$ Distance to Bonus	58.004***	17.217***
	(9.845)	(1.811)
Observations	626,187	626,187
Adjusted R^2	0.0001	

◂ Number of Trades

Bonus Achievement Difference in Differences

	Returns	Number of Trades	Volatility
	(1)	(2)	(3)
Day -5	-0.001**	-0.095**	0.003
	(0.0002)	(0.045)	(0.002)
Day -4	-0.001***	-0.113**	0.004
	(0.0002)	(0.052)	(0.002)
Day -3	-0.001**	-0.163***	0.006
	(0.0003)	(0.061)	(0.003)
Day -2	-0.002**	-0.143	-0.003
	(0.001)	(0.103)	(0.004)
Day 0	0.016***	0.051	0.004
	(0.001)	(0.127)	(0.004)
Day +1	-0.006***	-0.091	0.001
	(0.0004)	(0.073)	(0.003)
Day +2	-0.002***	-0.063	0.0002
	(0.0003)	(0.053)	(0.002)
Day +3	-0.001***	-0.064	0.001
	(0.0002)	(0.051)	(0.003)
Day +4	-0.001***	-0.074	-0.001
	(0.0002)	(0.044)	(0.002)
Day +5	-0.001***	-0.059	-0.001
	(0.0002)	(0.039)	(0.002)
Observations	3,250,182	3,250,182	3,250,182

◂ Returns ◂ Number of Trades

Transactions When Bonus Achieved: Sales

Dependent Variables:	Next 10 Day Returns		Previous 10 Day Returns	
	(1)	(2)	(5)	(6)
Bonus Achieved	0.012***	0.007***	0.045***	0.021***
	(0.001)	(0.001)	(0.001)	(0.001)
Sale Transaction	-0.002*	-0.001	0.004**	0.005***
	(0.001)	(0.001)	(0.001)	(0.001)
Bonus Achieved × Sale Transaction	0.005***	0.004***	0.022***	0.020***
	(0.001)	(0.001)	(0.001)	(0.001)
Previous 10 Day Returns	0.129***	0.121***		
	(0.003)	(0.003)		
Daily Return			0.022***	0.020***
			(0.005)	(0.004)
Fixed-effects				
Fund	Yes	Yes	Yes	Yes
Date		Yes		Yes
Observations	3,620,016	3,620,016	3,635,974	3,635,974
Adjusted R^2	0.026	0.021	0.025	0.011

◂ Bonus Vs Non Bonus Days ◂ Purchases Vs Sales

Transactions When Bonus Achieved: Purchases

Dependent Variables:	Next 10 Day Returns		Previous 10 Day Returns	
	(3)	(4)	(7)	(8)
Bonus Achieved	0.017***	0.011***	0.067***	0.041***
	(0.001)	(0.001)	(0.002)	(0.001)
Buy Transaction	0.002*	0.000	-0.004**	-0.005***
	(0.001)	(0.001)	(0.001)	(0.001)
Bonus Achieved × Buy Transaction	-0.005***	-0.004***	-0.022***	-0.020***
	(0.001)	(0.001)	(0.001)	(0.001)
Previous 10 Day Returns	0.129***	0.121***		
	(0.003)	(0.003)		
Daily Return			0.022***	0.020***
			(0.005)	(0.004)
Fixed-effects				
Fund	Yes	Yes	Yes	Yes
Date		Yes		Yes
Observations	3,620,016	3,620,016	3,635,974	3,635,974
Adjusted R^2	0.026	0.021	0.025	0.011

◂ Bonus Vs Non Bonus Days ◂ Purchases Vs Sales